L7.1: Equivalence and similarity of matrices

Transcript Language

English

Hello, and welcome to the match 2 component of the online degree course on data science.

In this video, we are going to talk about equivalence and similarity of matrices.

So, we want to impose certain equivalence or what is called an equivalence relation

based on two different relations, we are going to study this on the set of matrices.

So, the first one is called equivalence, this is the set of m by matrices.

And then the second one is for square matrices. So, the reason we want to do this is because there

are certain what I will call invariants. This may not make sense right now, but that is fine

called Eigen values, that you are going to see later on and

these will come up not in this course, but in the data science part that you will see

in the subsequent semesters where you will use these for certain calculations. So,

without bothering about what everything that I said before, whether or not it made sense, let us

get into the integrities of what is equivalence and similar similarity for matrices.

So, let us consider two, we will first talk about equivalence of matrices.

Let us consider two matrices of order m by n, both have the same order, remember.

So, we say A is equivalent to B, if B is equal to Q times A times B, so we can find matrices,

Q and P, which are invertible. So, P has to be because it is invertible, because I impose

the condition of being invertible. First of all, they have to be square matrices.

And since we are seeing QAP that means A is remember m by n, that means P has to be n by n,

and Q has to be m by m, so they have to be invertible matrices,

and B has to be equal to Q times A times P. So, you have, if we can find Q and P or P and Q,

such that B is equal to Q times A times P meaning when you compute Q times A times P,

you get B, then we say A and B are equivalent, or A is equivalent to B.

So, the other way of viewing this is by doing our good old row and column operations.

So, if you can perform a series of row operations on A, and then a series of column operations on A,

so that you can then obtain B, then we will say that A is equal to B. So, you take all the

column operations on the right,

so you get P and you take all the row operations on the left, and that is you get Q

by multiplying successively that is. So, we have seen this idea before.

And I am just trying to motivate how to get Q and P, if you want to actually get it.

But we will see a different way of getting them. The other characterization is that the

rank is the same. So, the reason this works is because remember that if you allow both

row and column operations then you can take any matrix and reduce it to the form where you have

block matrices, where you have the identity matrix 0,0,0. I will say that carefully in a second.

So, equivalence of matrices is an equivalence relation. So, what is an equivalence relation? So,

I am not defining it, but what it means is, A is equivalent to itself.

Remember that we say A is equal to B if there exists

P and Q, so that B is equal to P time Q times A times P. So, A is equivalent to itself is

saying that there exists some Q and P, so that A is equal to Q times A times P.

Now, I think it is clear what Q and P should be. We will write it down in a second.

So, A is equivalent to B implies B is equivalent to A. Again, because of invertibility of P and Q,

this is clear and A is equal to B and B to C implies A is equivalent to C. So, let us quickly write down how to do this. A is equivalent to itself because you can write A as

identity m cross m times A times identity n cross n, of course, identity matrices are invertible.

A is equivalent to B implies B is equivalent A. Well, we know that A is equivalent to B

that means B is equal to P times sorry, Q times

A times P, and that implies by multiplying by the inverses that A is equal to

Q inverse times B times P inverse. So, A is equivalent to B,

that is what we started with that means B is QAP, that will imply that B is equivalent to A.

So, because we can write A as Q inverse times B times P inverse. So, that means

you can multiply on the left by some invertible matrix and on the right by some invertible matrix

to get A from B, so B is equivalent to A. And then A is equal to B and B to C implies A is

equivalent to C. This is because B is equal to QAP, C is equal to let us say, Q prime,

B, P prime, then you can replace B by the first part. So, C is equal to

Q prime times Q times A times P times P prime. And now Q and Q prime are invertible,

that means this product is invertible. Similarly, P and P prime are invertible,

so this product is invertible. So, I have an invertible matrix on the left and on the right,

which I can multiply to A, in order to get C so that is why A is equivalent to C. And

now we can just say A and B are equivalent because this is an equivalence relation.

So, once you know that A is equivalent to B, B is also equivalent to A, so we can just say

A and B are equivalent. So, from now on, I may sometimes say A and B are equivalent matrices,

that means A is equivalent to B or B is equal to A both are the same. Now, let me come back

to this rank of A is rank of B. What I meant here was that

you can get Q and P so that QAP is of this form.

And you can also get Q prime and P prime so that Q prime B, P prime is of this form. And then

that is why A is equivalent to this matrix here if we call it C, and B is equal to this matrix C, but this is an equivalence relation, that means A is equivalent to B. So, that is one

way of checking that if rank of A is rank of B, then indeed, they are equivalent. Conversely, if

they are equivalent then I will leave it to you to check that indeed,

the ranks are the same that is a little bit of manipulation of

column and row spaces. So, we know what it means to say that two matrices are equivalent.

So, let us look at the linear transformation f from R3 to R2, given by f of x, y, z is x plus y,

and y plus z. So, let us consider two ordered bases for R3, the standard ordered basis, and

another ordered bases which is 1, 1, 0, 0, 1, 1 and 0, 0, 1. And let us consider

two ordered bases for R2, which is a standard ordered bases and the bases, 1, 0 and 1,1.

So, let us express the standard ordered bases for R3 in terms of the standard ordered basis

for R2 after applying f. So, if you apply f to 1, 0, 0 you get

1 plus 0, comma 0 plus 0, so that gives you 1, 0. If you apply f to 0, 1, 0 you get

0 plus 1 and 1 plus 0, which is 1, 1 and then you write it in terms of the standard bases, which is

one times 1, 0 plus one times 0, 1. And then if you apply f to 0, 0, 1 you get 0 plus 0 and 0 plus

1, which is 0, 1 and that is just one times the second vector in the ordered standard bases.

So, if you want to write down the corresponding matrix for this linear transformation, that is

corresponding to the ordered bases, standard ordered bases for R3 and standard, ordered

basis for R2 you get 1,0, 1,1 and 0,1. So where is this 1,0 coming from it is coming from here.

Where is this 1,1 the second column coming from, it is coming from these coefficients. And the

third one is coming from these coefficients. This is something we did in our previous videos.

Let us do the same thing for the other two bases. So, the other standard, the other ordered bases

for R3 and the other ordered bases for R2. So, if you look at f of 1,1,0 that is 1 plus 1 comma 1

plus 0, that is 2,1, which you can express as one times 1,0 plus one times 1,1. In these the ordered bases the second ordered bases gamma 2. Similarly, if you take f of 0,1,1 this is

0 plus 1 comma 1 plus 1, which is 1 comma 2. And for this, if you order, if you write it in

terms of the ordered basis it gives you minus one times 1 comma 0 plus two times 1 comma 1.

And for the last one, you have f of 0,0,1, which is 0 plus 0 comma 0 plus 1, which is 0,1, which

is minus 1 times 1,0, plus one times 1,1. So, if you look at the corresponding matrix,

first column is going to come from the first equation coefficients, the second column is

going to come from the coefficients of the second equation and the third column is going to come

from the coefficients of the third equation. So, it will look like 1,1 minus 1,2 and minus 1,1.

So, what have we done so far, we have taken a linear transformation and we have written

down two matrices corresponding to two different ordered sets of ordered bases.

So, choose P to be this matrix 1, 0, 0 and 1, 1, 0, 0 ,1, 1 and Q to be 1 minus 1,0,1. So, I, this is something I produced out of the air and here is what happens. So then,

Q times A times P, this is something you have to compute, I have done the computation here.

This is 1 minus 1, 0, 1 times 1, 1, 0, 0, 1, 1. So, this was A times 1, 0, 0, 1, 1, 0, 0, 1, 1.

So, if you multiply this out, let us see what you get after doing the first multiplication, you get

the first row in out of out of these two is, so 1 minus one times so this is 1, then this is 0,

then this is minus 1, and then you have 0,1,1. And you want to multiply this to

this matrix. So, if you do that, let us see what we get. So, if you do 1, 0 minus 1,

indeed, you get 1, then 1, 0 minus one times 0,

1, 1 that is minus 1, and similarly a minus 1. So, this first row is indeed achieved and then

for the second row, you take 0, 1, 1 and multiply it over there so that gives you 1, 2 and 1, so

indeed the second row is achieved. So, if you do this multiplication yourself, you will be able to

see that Q times A times P is indeed equal to B. So, these matrices are equivalent to each other.

So, this is not some a magic. So, consider a linear transformation T,

V to W, and two ordered bases beta 1 and beta 2 for V, and two ordered bases gamma 1 and gamma

2 for W. So, like in the previous example, where we had V was R3, W was R2 and beta 1 and gamma 1

with a corresponding standard ordered bases beta 2 and gamma 2 are some other fixed ordered basis.

So, we took A to be the matrix corresponding to beta 1 and gamma 1.

So, similarly, here, you take A to be the matrix corresponding to T with respect to the basis

beta 1 gamma 1 and you take B to be the matrix corresponding to T with respect to the basis beta

2 and gamma 2, this is exactly what we did in the previous example, then A is equal to B.

So, the previous example is an example of this phenomenon that is that we have mentioned here.

So, the question is, how do I get Q and how do I get P? So, for P, express the

bases beta 2 in terms of beta 1, and for Q express

the bases, I should say ordered bases, everywhere we have ordered bases, the ordered bases gamma 1 in terms of gamma 2. So, this is how I got those two matrices.

Let us quickly look at that example again. So, right. So, let us look at this example again.

So, if you look at the matrix P, the matrix P is exactly expressing this basis here 1, 1, 0

so this is one times E1 plus E2 plus 0 times E3. So, 1, 1, 0 goes into your first column,

and similarly each of these, so you have 0, 1, 1 which goes into your second column and 0, 0,

1 which goes into your third column, and so that is how I get P.

And for Q, what I did was, I wrote 1, 0 and 0, 1 in terms of 1, 1 and

sorry, in terms of 1, 0 and 1, 1 so this is one times 1, 0 plus 0 times 1, 1.

That is why I get 1, 0 in my first column and 0, 1 is

minus one times 1, 0 plus one times 1, 1 that is how I get minus 1,1.

So, this is a this is a strategy for getting P and Q and this works for any

general linear transformation and these fixed ordered basis beta 1 beta 2 and gamma 1, gamma 2.

So, yeah, so to complete this express this, get P and Q from there and then you will find that

B is equal to QAP. So, this is something you have to check. I will leave it at that since

this is not really a proof-based course. So, we have exhausted equivalence, let us

look at similarity of matrices. So now, we are going to restrict ourselves to square matrices.

So, we will say that two square matrices of size n by n are similar or rather we will say that A is

similar to B. So, we have two n by n matrices A and B and we will say A is similar to B,

if there is an invertible matrix P, it has to be of size n by n such that B is

equal to P inverse AP. So, now the role of Q is being played by P inverse. So, this is a very

big change. So, if you can express B as P inverse AP, then we will say that A is similar to B.

So, again similarities and equivalence relation by which we mean A is similar to itself. A similar

to B implies be similar to A and A similar to B and B to C implies A similar to C. Let us quickly see why this is the case. So, for the first one, again, you can take P to be identity,

then A is equal to identity inverse A identity. Well, identity inversus identity, so this is just

identity times A times identity, which is certainly A. For the second one. Let us see

how to do the second one, these are on the same lines that we have done the previous part.

So, A similar to B means B is equal to P inverse AP, but that means, because P is invertible,

I can multiply by P inverse on the right and P on the left, so I get PBP inverse is equal to A. So,

just rewriting that P is equal to P times, so I want to write this as P inverse,

inverse. So, why did I write it like that, because I want to express this in terms of

the inverse of whatever is occurring on the right term, P inverse times B times P inverse.

Now, you take your P inverse, call it something else call it P prime, if you want, then we have

an invertible matrix P prime such that A is equal to P prime inverse times B times P prime, which is exactly saying that B is similar to A. So, the second one is satisfied. And

the third one again on the same lines as we did our previous part. So, let us see.

So, we have these B is P inverse AP and C is let us say Q inverse

AQ, these are both n by n invertible matrices P and Q. So, then C is equal to

my bad, this is not B, but this is this is not A but this is B. So, C is equal to Q inverse and now

you put in the values put in the first equation, which is B is B inverse AP into the second,

then you get Q inverse P inverse times A times PQ, but remember Q inverse P inverse is exactly

PQ inverse. And now if I call P times Q by something else, let us say P prime,

then this is saying that C is equal to P prime inverse times A times P prime,

which says that A is similar to C. So, this proves the three things. So,

this says that similarity is what is called an equivalence relation. So, now I can say,

especially because of the second one, instead of saying that A is similar to B I can just say A and

B are similar because it does not matter which is similar to which the other one always follows. So,

A is similar to B, can be replaced by saying A and B are similar matrices.

So, why do we really care about similar matrices?

So, the reason we care about similar matrices is because of some of these properties. The

first is that A and B are equivalent, this is clear, because equivalent means

that there is P and Q so that B is Q times A times P, where P and Q are invertible,

but if they are similar then you can take Q to be P inverse, and P to be P. So, this is clear. A and

B have the same rank as a result, the determinant is the same. Now, this is a big difference.

Remember, now we are in square matrices, we could not have made any such

kind of statement before. The determinant of B is determinant of P inverse times AP,

but determinants multiply, if you remember, then that means this is determinant of P inverse times

determinant of A times determinant of P. But now, although, matrices do not commute real numbers do, and determinants give you real numbers. So, I can write this as

1 by determinant of P, because determinant of P inverse is 1 by determinant of P

times determinant of A times determinant of P, and then because these are real numbers,

I can shift 1 by determinant of P to the right, and then determinant of P divided by determinant

of P gives me 1, so I get just determinant of A. So, they have the same determinants.

This is a very useful property and this is one of the properties that you will

be using later on. Not in this course, but in a different course for certain reasons.

So, the reason, in fact, is that there are other invariants, so one such as called the

Eigen values which you are going to study later. So, I have written some names here. And this,

you are not expected to make sense of this, but it is good, it will be good if you remember these,

this just as a statement without understanding when you do your next course.

So, several other invariants of A and B are the same such as the characteristic polynomial, the minimal polynomial and the Eigen values with multiplicity. We are not

going to define this right now, but this is just as a feeder into the next course,

where you will be using some of these things, especially, this thing called the Eigen values.

So, just remember this name. Does not matter if you do not understand what it is now. So, this is

why we are studying this idea of similarity. Let us do some examples or maybe one example.

So, suppose you have a linear transformation f of x, y, z is minus x plus y plus z, x minus y plus

z and x plus y minus z. Now, we are going to use the same standard the same bases. Earlier, we had

two different bases. So, we had beta 1 and gamma 1, and then we had beta 2 and gamma 2. But here,

remember that this linear transformation is from R3 to R3 itself. So, we are going to use

the same ordered bases. So, let us start with beta being gamma being the standard ordered basis.

So, let us do what we did before. Let us express f of 1,0,0, f of 0,1,0, and f of ,0,0,1 in terms of the standard basis. If you do that, you get f of 1,0,0 is minus 1,1,1,

f of 0,1,0 is 1 minus 1,1 and f of 0,0,1 is 1,1 minus 1.

So that means these things exactly end up being the coefficients, when you write it

in terms of the standard ordered bases. And as a result, the matrix of f corresponding to these,

the standard ordered bases is coming from these give you your first column these coefficients

give you’re your second column and these coefficients give you your third column.

Let us choose some other bases, beta prime and gamma prime is the same. So, this is the same,

so this is beta prime, and it is also gamma prime. So, let us express this f of

each of these vectors in terms of as linear combinations of these vectors themselves.

So, if I do f of 1,1,1, I get back 1,1,1. Let us remember what the linear transformation was it was

minus x plus y plus z, x minus y plus z and x plus y minus z.

So, in that case we get f of 1,1,1 is minus 1 plus 1 plus 1, 1 minus 1 plus 1, 1 plus 1 minus 1 which is 1,1,1, so which is the first factor itself. So, this is one times 1,1,1 plus 0 for the other 0

is in the coefficients for the other vectors. f of minus 1,1,0 is minus of minus 1, which is 1,

plus 1, plus 0, which is 2, then minus 1 minus 1 plus 0, which is minus 2,

and then 1 minus 1, rather minus 1 plus 1 plus 0, which is 0, so you get 2 minus 2, 0

and f of minus 1,0 comma 1 is, again, you will, if you do it you will get 2,0 minus 2.

And now if you write these in terms of the vectors that you have, the second 2 minus 2,0

is just two times the second vector or other minus two times the second vector.

And the f of the third vector is minus two times the third vector. So, what

we have got is, there is a basis here, which is beta prime, so that when I apply f on that basis

for these particular vectors in the bases, they just scale.

So, f of 1,1,1 is 1,1,1, f of minus 1,1,0 is minus 2 times minus 1,1,0 and 4 minus 1,0,1 is

minus two times minus 1,0,1. So, for these particular bases, the linear transformation behaves very nicely, it just scales. One of the vectors remains the

same so it scales by 1, the second and third vector get scaled by minus 2.

So, the corresponding matrix is diagonal matrix 1, 0, 0, 0 minus 2, 0 and 0, .0 minus 2.

And now, we know that diagonal matrices are particularly easy to understand. So,

what we see here is that, we have the same linear transformation in terms of one basis

the standard ordered bases, we got a matrix which was not hard, but not maybe not so easy.

But in terms of this bases, this new basis beta prime

ordered bases, we saw that it is a diagonal matrix.

(Refer Slide Time: 27:17) Now, let P be 1 minus 1 minus 1,1,1,0,1,0,1

compute P inverse, this is something you can do, I have done it here. It is one third, one third,

one third, minus one third, two third minus one third, minus one third, minus one third, two

third. And now, if you multiply, so, you look at P inverse AP, then you get. If you multiply this carefully, I have done it here. So, I will suggest you check that this calculation is correct,

you get the matrix B. So, this last matrix is

exactly your diagonal matrix B. So, you get that A and B are similar matrices.

So, what have we seen, we have seen exactly what happened in equivalence, the same kind

of thing is happening for similarity, but this time, instead of having different bases,

your beta and gamma are the same ordered bases because your vector space V and is

the same vector space, it is V and V, so you have a linear transformations from V to V.

So, here we have R3 to R3. So, you take the bases, beta and beta. In our case, we do identity, sorry,

the standard ordered bases and you got the matrix A, which is this matrix here.

And in terms of a different ordered basis, which we call beta prime, where implicitly gamma prime

was equal to beta prime, we got this matrix. So, remember that now the basis that you are choosing

on both sides are the same. So, the beta and gamma have to match. And then if you write down these matrices, they end up being similar matrices. So, this is a general phenomenon. Let us look at

another quick example. So, f of x, y is 2x comma y. So, if you look at the ordered bases 1,0 and

1,1 you can look at. So, f of 1,0 is just 2,0, f of 1,1 is 2,1 and you can write this in terms

of our bases that you have 1,0 and 1,1. So, these bases is if you want beta and gamma both.

And as a result, the corresponding matrix is the coefficients coming from here,

coefficients which go into the first column, coefficients from there, which go into the

first, second column, so it is 2,1,0,1. Instead, you can look at the standard ordered

basis. So, this is this time your beta prime and gamma prime, and it is clear it is 2x comma y.

So, f of 1,0 is 2,0 f of 00,1 is 0,1, so this is scaling your first vector by 2 and it is

returning your second vector as it is. So, this gives us a diagonal matrix 2,0,0,1.

So let P be 1,1,0,1 P inverse is 1 minus 1,0 you can compute this.

And now if you compute P inverse AP, this is a very simple calculation, which I have done here, you get your diagonal matrix, which is exactly the matrix B. So, this is exactly B, so the matrices

2,1,0,1 and 2,0,0,1 are similar. So, we have seen two examples where

we took linear transformations from two vector spaces to themselves

expressed each time these linear transformations as basis with the same basis on both sides,

and then we change the basis. Again, with the new basis on both sides,

we computed the matrix and we saw that these matrices are similar.

So, this is a general phenomenon, and the proof for this is exactly the same.

So, consider a linear transformation T, V to V and two ordered bases beta and gamma for V.

Let A be the corresponding matrix with respect to beta and B be the corresponding matrix with

respect to gamma, then A is similar to B. So, that means I have to produce P. How do I produce P?

To produce P you write, so express, so we want that B is P inverse

AP, so express, so for P you express the, you express gamma in terms of beta,

and P inverse is well you can get it from P by inverting or you can get it from B by expressing

beta in terms of gamma. What does this mean? This means you take each in an ordered way, you take.

So, this first one means you take the first vector of gamma, write it as a linear combination of

elements of the vectors from beta, take the second vector of gamma write it as a linear combination

of beta and so on, so you get that. So, if you have n many, you write all those n many

things, and then you get a matrix. For the first equation that you have put the coefficients into

the first column, for the second equation, put them into the second column and so on for the nth

equation, put them into the nth column, that is your matrix P. Same thing, but the opposite way,

that is your matrix P inverse or you can get it just by inverting P. And you will find that you

end up with B is P inverse AP. So, again, it is not a proof-based course, I would not prove this fact, but this is how you get P fine. Why do we care about similarity? I mean, this is

maybe the most important question. So, the reason is that, as we saw in the previous two examples,

sometimes. So, we hope, so, the underlined part is we hope, because under some basis,

we hope that the corresponding matrix is a diagonal matrix, which gives an easy geometric

understanding of the linear transformation. So, if you have a diagonal matrix, as I said it

saying that some particular vector scales, some other particular vector scales and some third

vector letting our three scales and give me how much it scales by

and these three form a basis. So, that is the intuitive picture you have in mind.

So, let us quickly recall what we have done this video. We have seen the notion of similarity and

equivalence of matrices. Equivalence was, both of these are equivalence relations.

Equivalence is for rectangular matrices, I mean, which means matrices which need not be square.

So, we say they are equivalent to B is QAP, then A and B are equivalent.

For similarity, we say that B is P inverse AP, I mean, if these B is P inverse AP then

we say A and B are similar. And these notions are you are useful because they sort of help you to

understand how to go between different basis on the vector space if you want if you have

a particular linear transformation and you choose different basis, how do you go between

the corresponding matrices that is really what equivalence and similarities telling you.

And similarity is particularly important because, it may so, happen that your linear transformation

may not, the algebra that when you write it down, it may look like a very difficult

entity, but under some particular basis, it may suddenly start looking very nice. This

is really the main point behind why we study similar. So, thank you for your attention.